Pricing and hedging financial derivatives: PDE approach and Black-Scholes formula

Peter Tankov

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Bird's eye view and key concepts of this lecture

- *A Markov diffusion model for the financial market*
- *The pricing equation; explicit form of the hedging portfolio*
- *Black-Scholes formula and the greeks*
- *Beyond Black-Scholes: discretization error and gamma hedging; transaction costs via Leland's approach; volatility risk and robustness of the Black-Scholes formula; implied volatility*

PDE approach to pricing

- The risk-neutral pricing formula

$$V_t(G) = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r(s)ds} G | \mathcal{F}_t\right]$$

  does not allow to compute the hedging portfolio.
- The “standard” way to obtain the hedge ratio is through Itô formula in the Markovian setting.
- The Markovian setting also allows to express the option price as solution to a PDE.

Reminder: Stochastic differential equations

We consider an SDE:

$$dX_t = b(t, X_t)dt + a(t, X_t)dW_t, \quad X_0 = x \tag{$*$}$$

where W is a d-dimensional BM on $(\Omega, \mathcal{F}, \mathbb{P})$ with filtration $(\mathcal{F}_t)_{t\geq 0}$,
$a : [0, T] \times \mathbb{R}^n \mapsto \mathbb{R}^{n\times d}$ and $b : [0, T] \times \mathbb{R}^n \mapsto \mathbb{R}^n$.

A strong solution of $(*)$ is a $(\mathcal{F}_t)_{t\geq 0}$-adapted process X which satisfies $(*)$ and

$$\int_0^T |b(s, X_s)|ds + \int_0^T |a(s, X_s)|^2 ds < \infty.$$

Assume that $\exists K > 0$ such that $\forall t \in [0, T]$, $\forall x, y \in \mathbb{R}$,

$$|a(t,x)|+|b(t,x)| \leq K(1+|x|) \text{ and } |a(t,x)-\sigma(t,y)|+|b(t,x)-b(t,y)| \leq K|x-y|.$$

Then, $(*)$ admits a unique strong solution with finite moments.

The diffusion model for a financial market

Let the prices of risky assets be given by the solution of the system of SDE:

$$dS_t = \mathrm{diag}[S_t](\mu(t, S_t)dt + \sigma(t, S_t)dW_t),$$

and the risk-free asset be given by

$$S_t^0 = \exp\left(\int_0^t r(u, S_u)du\right).$$

We assume that $\mu \in \mathbb{R}^d$, $\sigma \in \mathbb{R}^{d\times d}$, σ^{-1} and $r \geq 0$ are bounded; $\mathrm{diag}[S]\mu$ and $\mathrm{diag}[S]\sigma$ are Lipschitz continuous in S uniformly in t.

$\Rightarrow$ under these assumptions, the SDE admits a unique strong solution, and the risk premium $\lambda(t, S) := \sigma(t, S)^{-1}(\mu(t, S) - r(t, S)\mathbf{1})$ is bounded.

Risk-neutral probability

Introduce the risk-neutral probability $\mathbb{Q}$ via

$$\frac{d\mathbb{Q}}{d\mathbb{P}}\Big|_{\mathcal{F}_T} = \exp\Big(-\int_0^T \lambda(t,s)dW_t - \frac{1}{2}\int_0^T \|\lambda(t,s)\|^2 dt\Big).$$

Then,

$$\widehat{W}_t = W_t + \int_0^t \lambda(t,s)ds$$

is a $\mathbb{Q}$-Brownian motion, the prices of risky assets satisfy

$$dS_t = \mathrm{diag}[S_t](r(t,S_t)\mathbf{1}dt + \sigma(t,S_t)d\widehat{W}_t),$$

and the discounted prices satisfy

$$d\widetilde{S}_t = \mathrm{diag}[\widetilde{S}_t]\sigma(t,S_t)d\widehat{W}_t.$$

Burkholder-Davis-Gundy inequalities and Gronwall inequality

- For any $p > 0$, there exist universal constants c_p and C_p such that for every continuous local martingale X,

$$c_p \mathbb{E}[\langle X \rangle_T^{p/2}] \leq \mathbb{E}[(X_T^*)^p] \leq C_p \mathbb{E}[\langle X \rangle_T^{p/2}],$$

  where $X_T^* := \max_{0 \leq t \leq T} |X_t|$.

- If u is a finite function and for some constants $\alpha \in \mathbb{R}$ and $\beta \geq 0$,

$$u(t) \leq \alpha + \beta \int_a^t u(s) ds, \quad t \in [a, b],$$

  then

$$u(t) \leq \alpha e^{\beta(t-a)}, \quad t \in [a, b].$$

Bounded moments

Proposition

For all $n \geq 1$,

$$\mathbb{E}^{\mathbb{Q}}\left[\max_{0\leq t\leq T} \|\widetilde{S}_t\|^n\right] < \infty.$$

To simplify notation, let $d = 1$ and $n \geq 2$. Let $\tau_M := \inf\{t > 0 : \widetilde{S}_t > M\}$ and $\widetilde{S}^M_t := \widetilde{S}_{t\wedge\tau_M}$. By Burkholder-Davis-Gundy, the bound on σ, and Jensen's inequality

$$\mathbb{E}^{\mathbb{Q}}[\max_{0\leq t\leq T} |\widetilde{S}^M_t - \widetilde{S}_0|^n] \leq C\mathbb{E}^{\mathbb{Q}}[\langle \widetilde{S}\rangle^{n/2}_{\tau_M}] = C\mathbb{E}^{\mathbb{Q}}\left[\left(\int_0^{\tau_M} \widetilde{S}^2_t \sigma^2(t, S_t)dt\right)^{n/2}\right]$$
$$\leq C\mathbb{E}^{\mathbb{Q}}\left[\left(\int_0^{\tau_M} (\widetilde{S}^M_t)^2 dt\right)^{n/2}\right] \leq C\mathbb{E}^{\mathbb{Q}}\left[\int_0^T (\widetilde{S}^M_t)^n dt\right]$$

where the constant C may vary from line to line.

Bounded moments: proof

Using the inequality $|a+b|^n \leq 2^n(|a|^n + |b|^n)$,

$$\mathbb{E}^{\mathbb{Q}}[\max_{0\leq t\leq T} |\widetilde{S}_t^M|^n] \leq C\widetilde{S}_0^n + C\int_0^T \mathbb{E}^{\mathbb{Q}}\left[(\widetilde{S}_t^M)^n\right] dt$$

and thus, by Gronwall's inequality,

$$\mathbb{E}^{\mathbb{Q}}[\max_{0\leq t\leq T} |\widetilde{S}_t^M|^n] \leq C\widetilde{S}_0^n e^{C'T}$$

Finally, monotone convergence allows to conclude that

$$\mathbb{E}^{\mathbb{Q}}[\max_{0\leq t\leq T} |\widetilde{S}_t|^n] < \infty.$$

Infinitesimal generator

Let $\mathcal{L}$ be the infinitesimal generator of the SDE:

$$\mathcal{L}_s u(t,S) = r(t,S)\sum_{i=1}^d S^i \frac{\partial u}{\partial S^i} + \frac{1}{2}\sum_{i,j=1}^d a_{ij}(t,S) S^i S^j \frac{\partial^2 u}{\partial S^i \partial S^j}$$

$$a_{ij}(t,S) = \sum_{k=1}^d \sigma_{ik}(t,S)\sigma_{jk}(t,S).$$

Then, for any $C^{1,2}$ function u, the process

$$M_t^u = e^{-\int_0^t r(s,S_s)ds} u(t,S_t) - \int_0^t e^{-\int_0^s r(v,S_v)dv}\left(\frac{\partial u}{\partial t} + L_s u - r(s,S_s)u\right) ds$$

$$= u(0,S_0) + \int_0^t e^{-\int_0^s r(v,S_v)} \sum_{i,j=1}^d \frac{\partial u}{\partial S^i} S_t^i \sigma_{ij}(s,S_s) dW_s^j$$

is a local martingale.

The pricing equation

Let that the pay-off at time T by given by $G = g(S_T)$, where $g \in C^0((0,\infty))$ with polynomial growth, and consider the pricing equation:

$$\frac{\partial u}{\partial t} + L_s u - r(t,S)u = 0, \quad (t,x) \in [0,T) \times (0,\infty),$$

with terminal condition $u(T,x) = g(x)$ for $x \in (0,\infty)$.

Assume that this equation admits a solution $u \in C^0([0,T] \times (0,\infty)) \cap C^{1,2}([0,T) \times (0,\infty))$ with polynomial growth in x uniformly on $t \in [0,T]$.

The unique price of the option is given by

$$V_t = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r(s,S_s)ds} g(S_T) \middle| \mathcal{F}_t\right] = u(t,S_t).$$

The pricing equation: one-dimensional case

In the one-dimensional case, pricing equation for European options takes the familiar form:

$$\frac{\partial u}{\partial t} + \frac{1}{2}\sigma^2(t,S)S^2\frac{\partial^2 u}{\partial S^2} = r(t,S)\left(u - S\frac{\partial u}{\partial S}\right), \quad (t,x) \in [0,T) \times (0,\infty).$$

This is known as Black-Scholes PDE.

The pricing equation: proof

Let u be the solution of the pricing equation, and define the process M^u as above:

$$M_t^u = e^{-\int_0^t r(s,S_s)ds} u(t,S_t) = u(0,S_0) + \int_0^t e^{-\int_0^s r(v,S_v)} \sum_{i,j=1}^d \frac{\partial u}{\partial S^i} S_t^i \sigma_{ij}(s,S_s) dW_s^j$$

Define a sequence of stopping times

$$\tau_n = \inf\{t > 0 : |\log S_t/S_0| > n\} \wedge T - \frac{1}{n}$$

Then, $\tau_n \to T$ as $n \to \infty$ and for every n, M^u stopped at τ_n is a square integrable martingale so that $M^u_{\tau_n \wedge t} = \mathbb{E}^{\mathbb{Q}}\left[M^u_{\tau_n}\middle|\mathcal{F}_t\right]$ and

$$e^{-\int_0^{t\wedge\tau_n} r(s,S_s)ds} u(t\wedge\tau_n, S_{t\wedge\tau_n}) = \mathbb{E}^{\mathbb{Q}}[e^{-\int_0^{\tau_n} r(s,S_s)ds} u(\tau_n, S_{\tau_n})|\mathcal{F}_t]$$

Taking the limit $n \to \infty$, using the dominated convergence theorem with the moment bound, we get:

$$u(t,S_t) = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_t^T r(s,S_s)ds} g(S_T)\middle|\mathcal{F}_t\right].$$

The hedging portfolio

Consider a portfolio with initial value $x = u(0, S_0)$ and strategy

$$\pi^i_t = \frac{\partial u}{\partial S^i} S^i_t, \quad t < T \quad \text{and} \quad \pi^i_T = 0.$$

For $t < T$, since $\frac{\partial u}{\partial S}$ is continuous, $\int_0^t \|\tilde{\pi}_s \sigma(s, S_s)\|^2 ds < \infty$ p.s. and, by Itô formula,

$$X^x_t = u(t, S_t) \quad \Longleftrightarrow \quad \widetilde{X}^x_t = \mathbb{E}^{\mathbb{Q}}\left[e^{-\int_0^T r(s,S_s)ds} g(S_T) \middle| \mathcal{F}_t \right],$$

so that

$$\mathbb{E}^{\mathbb{Q}}[\langle \widetilde{X}^x \rangle_t] = \mathbb{E}^{\mathbb{Q}}\left[\int_0^t \|\tilde{\pi}_s \sigma(s, S_s)\|^2 ds \right] = \mathbb{E}^{\mathbb{Q}}[(\widetilde{X}^x_t)^2 - x^2] < C < \infty,$$

and by monotone convergence, $\mathbb{E}^{\mathbb{Q}}\left[\int_0^T \|\tilde{\pi}_s \sigma(s, S_s)\|^2 ds \right] < \infty$, so that the portfolio is admissible on $[0, T]$.

Finally, by continuity, $X^x_T = g(S_T)$, the portfolio replicates the option.

Back to Black-Scholes: delta-hedging

The dynamic hedging strategy allowing to eliminate the risk associated to a Call option in the Black-Scholes model is given by

$$\delta_t = \frac{\partial \text{Call}}{\partial S}(t, S_t) = N(d_1), \qquad d_1 = \frac{\log \frac{S}{Ke^{-r(T-t)}} + \frac{\sigma^2(T-t)}{2}}{\sigma\sqrt{T-t}}.$$

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Proof:

$$\begin{aligned}\frac{\partial \text{Call}(t,S)}{\partial S} &= \frac{\partial}{\partial S}\mathbb{E}[e^{-r(T-t)}(Se^{(r-\frac{\sigma^2}{2})(T-t)+\sigma W_{T-t}} - K)^+]\\ &= \mathbb{E}[e^{-\frac{\sigma^2}{2}(T-t)+\sigma W_{T-t}}\mathbf{1}_{Se^{(r-\frac{\sigma^2}{2})(T-t)+\sigma W_{T-t}}\geq K}]\\ &= \widetilde{\mathbb{Q}}[Se^{(r+\frac{\sigma^2}{2})(T-t)+\sigma \widetilde{W}_{T-t}} \geq K] = N(d_1).\end{aligned}$$

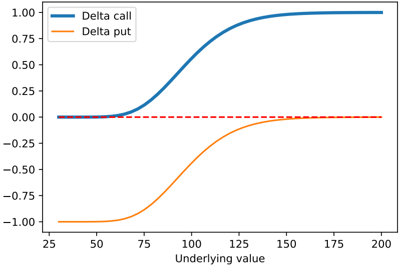
To understand the behavior of options as function of the model parameters / risk factors and quantify / manage risks associated to parameter variation, compute the sensitivities of Black-Scholes formula to these parameters.

The greeks: delta

The delta is the sensitivity to the present value of the underlying:

$$\frac{\partial C_{BS}}{\partial S} = N(d_1), \qquad \frac{\partial P_{BS}}{\partial S} = N(d_1) - 1.$$

It is by far the most important risk factor affecting the option price.

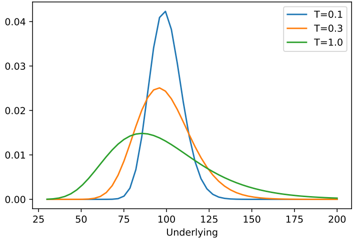


Greeks: gamma

The gamma is defined as the second derivative of the price or first derivative of delta.

$$\frac{\partial^2 C_{BS}}{\partial S^2} = \frac{\partial^2 P_{BS}}{\partial S^2} = \frac{\partial}{\partial S} N(d_1)$$

$$= \frac{n(d_1)}{S\sigma\sqrt{T-t}}, \quad n(x) = \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} = N'(x).$$

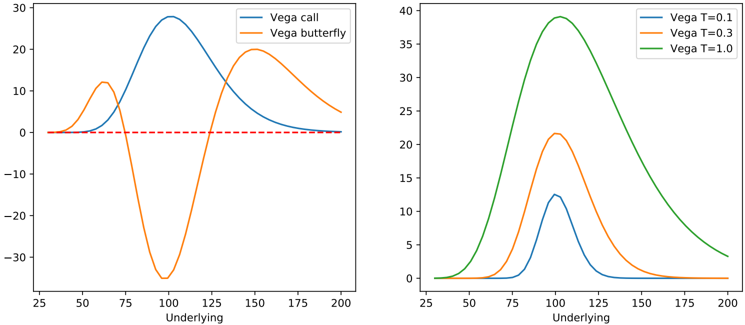


- Double interpretation of price sensitivity to large variations of underlying, and the rate of change in delta.
- Since gamma is always positive, call/put prices are convex in S.
- The gamma of an option is large when the option is close to expiry.

Greeks: vega

The vega is the sensitivity to the volatility:

$$\frac{\partial C_{BS}}{\partial \sigma} = \frac{\partial P_{BS}}{\partial \sigma} = Sn(d_1)\sqrt{T-t}.$$



Greeks: vega

The vega is the sensitivity to the volatility:

$$\frac{\partial C_{BS}}{\partial \sigma} = \frac{\partial P_{BS}}{\partial \sigma} = Sn(d_1)\sqrt{T-t}.$$

- Call and put prices are increasing in volatility.
- Vega is large at the money, but decreases for options close to expiry.
- A portfolio of n options with prices $C_1, \dots, C_n$ containing w_i units of i-th option is gamma-neutral (resp., vega-neutral) if

$$\sum_{i=1}^n w_i \frac{\partial^2 C_i}{\partial S^2} = 0 \quad \text{respectivement} \quad \sum_{i=1}^n w_i \frac{\partial C_i}{\partial \sigma} = 0.$$

- In Black-Scholes, if all options have the same expiry, vega-neutral equals gamma-neutral.

Other greeks

The rho is the sensitivity to interest rate:

$$\frac{\partial C_{BS}}{\partial r} = K(T-t)e^{-r(T-t)}N(d_2) > 0, \quad \frac{\partial P_{BS}}{\partial r} = -K(T-t)e^{-r(T-t)}N(-d_2) < 0.$$

The call in increasing and the put is decreasing as function of interest rate.

The theta measures the time dependence when all other variables are constant.

$$\frac{\partial C_{BS}}{\partial t} = -\frac{Sn(d_1)\sigma}{2\sqrt{T-t}} - rKe^{-r(T-t)}N(d_2) < 0$$
$$\frac{\partial P_{BS}}{\partial t} = -\frac{Sn(d_1)\sigma}{2\sqrt{T-t}} + rKe^{-r(T-t)}N(-d_2)$$

Call theta is always negative but put theta may be both positive and negative

Gamma hedging and discretization error

- Hedging portfolios are readjusted at discrete dates: either fixed (e.g., daily, at market close) or determined from option parameters
- The remaining risk is small in normal conditions but may be important if something happens in the market
- If the trader is long gamma (convexity), the discretization risk is limited

Taylor expansion of order 1 in t and order 2 in S, assuming $r = 0$:

$$u(T, S_T) - u(0, S_0) = \sum_{i=1}^{n} \{u(t_i, S_{t_i}) - u(t_{i-1}, S_{t_{i-1}})\}$$

$$\approx \sum_{i=1}^{n} \left\{ \frac{\partial u}{\partial t}(t_{i-1}, S_{t_{i-1}})\Delta t_i + \frac{\partial u}{\partial S}(t_{i-1}, S_{t_{i-1}})\Delta S_{t_i} + \frac{1}{2}\frac{\partial^2 u}{\partial S^2}(t_{i-1}, S_{t_{i-1}})\Delta S_{t_i}^2 \right\}$$

$$\sum_{i=1}^{n} \left\{ \frac{\partial u}{\partial t}\Delta t_i + \frac{\partial u}{\partial S}\Delta S_{t_i} + \frac{1}{2}\frac{\partial^2 u}{\partial S^2} S_{t_{i-1}}^2 \sigma^2 \Delta t_i + \frac{1}{2}\frac{\partial^2 u}{\partial S^2} S_{t_{i-1}}^2 \sigma^2 (\Delta W_{t_i}^2 - \Delta t_i) \right\}.$$

Gamma hedging and discretization error

Using the BS PDE: explicit form of the discretization error

$$\text{P\&L}_T = u(T, S_T) - X_T \approx \frac{1}{2}\sum_{i=1}^{n} \frac{\partial^2 u}{\partial S^2} S_{t_i-1}^2 \sigma^2 (\Delta W_{t_i}^2 - \Delta t_i),$$

or, with nonzero interest rate:

$$u(T, S_T) - X_T \approx \frac{e^{rT}}{2}\sum_{i=1}^{n} e^{-rt_{i-1}} \frac{\partial^2 u}{\partial S^2} S_{t_i-1}^2 \sigma^2 (\Delta W_{t_i}^2 - \Delta t_i),$$

The increment $\Delta W_{t_i}^2 - \Delta t_i$ has zero expectation and variance

$$\mathbb{V}ar\left[\Delta W_{t_i}^2 - \Delta t_i\right] = 2\Delta t_i^2.$$

By a functional central limit theorem, the process

$$\sum_{t_i \leq t} \frac{1}{\sqrt{2\Delta t_i}} \left(\Delta W_{t_i}^2 - \Delta t_i\right),$$

converges in law to a standard BM W^* independent from W.

Gamma hedging and discretization error

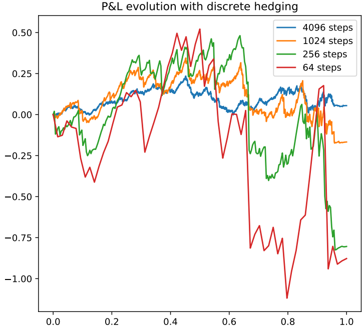
Finally,

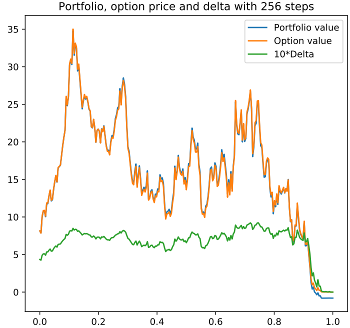
$$\text{P\&L}_T \approx \sqrt{\frac{\Delta t}{2}} e^{rT} \int_0^T e^{-rt} \frac{\partial^2 u}{\partial S^2} S_t^2 \sigma_t^2 dW_t^*.$$

- The discretization error is proportional to option's gamma and square root of time step.
- It is larger when gamma is higher: close to the money and to the expiry date.
- To reduce discretization error: increase readjustment frequency, or reduce gamma via gamma hedging, possible for large portfolios.

$$\gamma_t = \frac{\frac{\partial^2 u}{\partial S^2}}{\frac{\partial^2 u_0}{\partial S^2}}.$$

Gamma hedging and discretization error





Optimizing the readjustment frequency

Assuming a fixed cost δ per transaction and rebalancing frequency ω_t, on a small interval $[t, t+\Delta t]$ (still containing many readjustment dates) we get:

$$\text{Cost:} \quad \delta\omega_t \Delta t \qquad \text{Error:} \quad \sqrt{\frac{\Delta t}{2\omega_t}} \int_t^{t+\Delta t} \frac{\partial^2 u}{\partial S^2}(u, S_u)\sigma^2 S_u^2 dW_u^*$$

which leads to the optimal frequency

$$\omega_t = \left(\frac{\Gamma_t^2 S_t^4 \sigma^4}{4\delta^2}\right)^{\frac{1}{3}}.$$

The discretization error can be further reduced by readjusting the portfolio at stochastic dates (which are determined by the deviation of the portfolio from its initial value).

Transaction costs: Leland's approach

- The assumption of no transaction costs is in general false
- There are multiple transaction costs: bid-ask spreads, exchange fees
- Selling/buying large quantities of low liquid assets may be problematic
- For liquid assets / low trading frequencies transaction costs are small
- In the presence of proportional transaction costs, it is impossible to replicate an option exactly (Soner, Shreve, Cvitanic '95): the superhedging price is the price of the stock
- Leland '85: simple strategy to compensate transaction costs on average

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Option Pricing and Replication with Transactions Costs

HAYNE E. LELAND*

ABSTRACT

Transactions costs invalidate the Black-Scholes arbitrage argument for option pricing, since continuous revision implies infinite trading. Discrete revision using Black-Scholes deltas generates errors which are correlated with the market, and do not approach zero with more frequent revision when transactions costs are included. This paper develops a modified option replicating strategy which depends on the size of transactions costs and the frequency of revision. Hedging errors are uncorrelated with the market and approach zero with more frequent revision. The technique permits calculation of the transactions costs of option replication and provides bounds on option prices.

Transaction costs: Leland's approach

Main idea: Black-Scholes hedging with volatility $\bar\sigma$ higher than the true one. This creates a positive drift in the P&L which compensates transaction costs on average.

Assume the trader uses the Black-Scholes model with volatility $\bar\sigma$ and let $r = 0$. The value of trader's portfolio writes:

$$
\begin{aligned}
X_T &= u_{\bar\sigma}(0, S_0) + \sum_{i=1}^n \frac{\partial u_{\bar\sigma}}{\partial S}(t_{i-1}, S_{t_{i-1}})\Delta S_{t_i} \\
&\quad - k \sum_{i=1}^n \left| \frac{\partial u_{\bar\sigma}}{\partial S}(t_{i-1}, S_{t_{i-1}}) - \frac{\partial u_{\bar\sigma}}{\partial S}(t_i, S_{t_i}) \right| S_{t_i} \\
&\approx u_{\bar\sigma}(0, S_0) + \sum_{i=1}^n \frac{\partial u_{\bar\sigma}}{\partial S}(t_{i-1}, S_{t_{i-1}})\Delta S_{t_i} \\
&\quad - k \sum_{i=1}^n \frac{\partial^2 u_{\bar\sigma}}{\partial S^2}(t_{i-1}, S_{t_{i-1}}) S_{t_{i-1}} |\Delta S_{t_i}|
\end{aligned}
$$

Transaction costs: Leland's approach

Using the pricing equation, approximate P&L evolution writes:

$$\begin{aligned}\mathrm{P\&L}_T = X_T - g(S_T) &\approx u(0,S_0) - u(T,S_T) + \sum_{i=1}^{n} \frac{\partial u}{\partial S}\Delta S_{t_i} - k\sum \frac{\partial^2 u}{\partial S^2} S_{t_{i-1}}|\Delta S_{t_i}|\\ &\approx \frac{1}{2}\sum S_t^2 \frac{\partial^2 u}{\partial S^2}\sigma^2(\Delta W_t^2 - \Delta t) + \sum S_t^2 \frac{\partial^2 u}{\partial S^2}\left\{\frac{(\bar\sigma^2 - \sigma^2)\Delta t}{2} - k\sigma|\Delta W_t|\right\}\end{aligned}$$

To cancel the P&L on average, take

$$\bar\sigma^2 = \sigma^2 + \frac{2k\sigma\mathbb{E}[|\Delta W_t|]}{\Delta t} \quad \Rightarrow \quad \bar\sigma = \sigma\sqrt{1 + \sqrt{\frac{8}{\pi}}\frac{k}{\sigma\sqrt{\Delta t}}}$$

Transaction costs: Leland's approach

Under Leland's choice of volatility, residual error satisfies

$$\mathrm{P\&L}_T \approx \sqrt{\frac{\Delta t}{2}} \int_0^T S_t^2 \sigma^2 \Gamma_t dW_t^* + k\sqrt{1 - \frac{2}{\pi}} \int_0^T S_t^2 \sigma \Gamma_t dW_t',$$

where W^* and W' are independent Brownian motions.

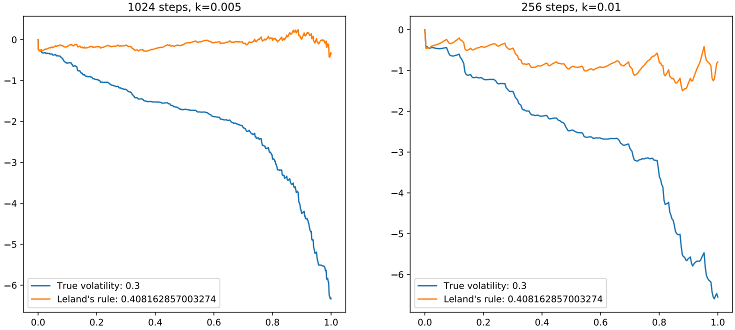
To make the additional error negligible as $k \to 0$, choose the discretization step such that $k = o(\sqrt{\Delta t})$.

To balance discretization error and the extra cost of hedging:

$$\frac{\Delta t}{2} \int_0^T S_t^4 \sigma^4 \Gamma_t^2 dt \quad \approx \quad \left(\frac{\partial u}{\partial \sigma}\right)^2 \frac{2}{\pi} \frac{k^2}{\Delta t}$$

The optimal discretization step is $\Delta t \sim k$.

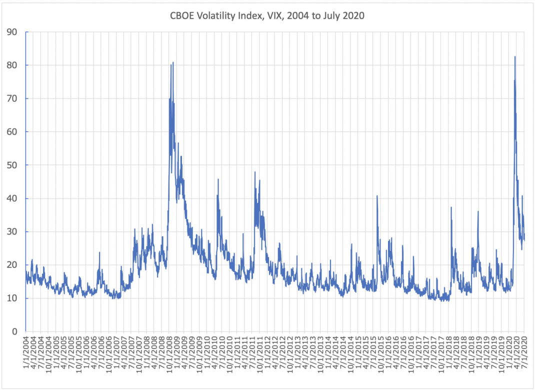
Transaction costs: Leland approach



Jupyter notebook of the lecture

Volatility risk and robustness of the BS formula

- The second most important risk after delta risk is the vega risk arising from changes in underlying volatility
- In practice volatility is stochastic: between 14/02/20 and 16/03/20, the VIX index went up from 13% to 82%; in normal times variations are also observed



Robustness of the Black-Scholes formula

Most important result in mathematical finance

Mathematical Finance, Vol. 8, No. 2 (April 1998), 93–126

ROBUSTNESS OF THE BLACK AND SCHOLES FORMULA

NICOLE EL KAROUI

Laboratoire de Probabilités, Université Pierre et Marie Curie

MONIQUE JEANBLANC-PICQUÉ

Equipe d'Analyse et Probabilités, Université d'Evry

STEVEN E. SHREVE

Department of Mathematical Sciences, Carnegie Mellon University

Consider an option on a stock whose volatility is unknown and stochastic. An agent assumes this volatility to be a specific function of time and the stock price, knowing that this assumption may result in a misspecification of the volatility. However, if the misspecified volatility dominates the true volatility, then the misspecified price of the option dominates its true price. Moreover, the option hedging strategy computed under the assumption of the misspecified volatility provides an almost sure one-sided hedge for the option under the true volatility. Analogous results hold if the true volatility dominates the misspecified volatility. These comparisons can fail, however, if the misspecified volatility is not assumed to be a function of time and the stock price. The positive results, which apply to both European and American options, are used to obtain a bound and hedge for Asian options.

KEY WORDS: option pricing, hedging strategies, stochastic volatility

Volatility risk and robustness of the BS formula

Assume true volatility σ_t

$$\frac{dS_t}{S_t} = \mu dt + \sigma_t dW_t,$$

but trader uses Black-Scholes model with constant volatility $\bar{\sigma}$, and let $r = 0$. Portfolio dynamics writes:

$$dX_t = \frac{\partial u_{\bar{\sigma}}}{\partial S}(t, S_t) dS_t, \quad X_0 = u_{\bar{\sigma}}(0, S_0).$$

Hedging error:

$$\varepsilon_T = X_T - g(S_T) = X_T - u_{\bar{\sigma}}(T, S_T).$$

Volatility risk and robustness of the BS formula

By Itô formula,

$$du_{\bar\sigma}(t,S_t) = \left(\frac{\partial u_{\bar\sigma}}{\partial t} + \frac{1}{2}\frac{\partial^2 u_{\bar\sigma}}{\partial S^2}\sigma_t^2 S_t^2\right)dt + \frac{\partial u_{\bar\sigma}}{\partial S}(t,S_t)dS_t,$$

but $u_{\bar\sigma}(t,S)$ as a function satisfies BS PDE:

$$\frac{\partial u_{\bar\sigma}}{\partial t} + \frac{1}{2}\bar\sigma^2 S^2 \frac{\partial^2 u_{\bar\sigma}}{\partial S^2} = 0.$$

Gathering three equations, we get:

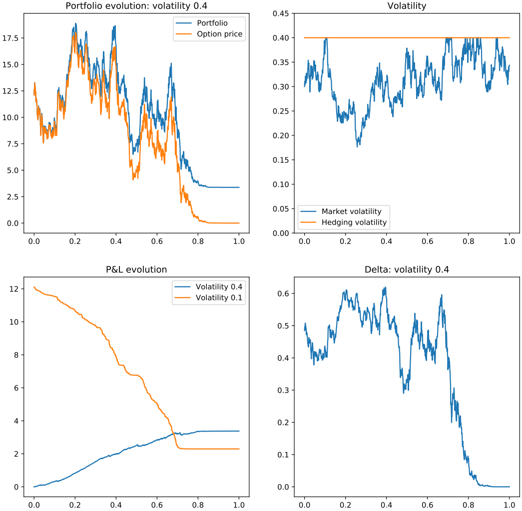
$$\varepsilon_T = \frac{1}{2}\int_0^T \frac{\partial^2 u_{\bar\sigma}}{\partial S^2}(\bar\sigma^2 - \sigma_t^2)S_t^2 dt.$$

or with nonzero interest rate,

$$\varepsilon_T = X_T - g(S_T) = \frac{1}{2}e^{rT}\int_0^T e^{-rt}\frac{\partial^2 u_{\bar\sigma}}{\partial S^2}(\bar\sigma^2 - \sigma_t^2)S_t^2 dt.$$

- If the true volatility if higher than $\bar\sigma$, P&L of option's buyer is positive
- If $\bar\sigma$ dominates the true volatility, seller has positive pay-off

Robustness of the BS formula: illustration



Implied volatility

In the BS model, unique unobservable parameter is the volatility.

The function $\sigma \mapsto C_{BS}(\sigma)$ satisfies

$$\lim_{\sigma \downarrow 0} C_{BS}(\sigma) = (S - Ke^{-r(T-t)})^+ \qquad \lim_{\sigma \uparrow \infty} C_{BS}(\sigma) = S$$
$$\frac{\partial C_{BS}}{\partial \sigma} = Sn(d_1)\sqrt{T-t} > 0.$$

Thus $C_{BS}(\sigma) = C$ has a unique solution for all C satisfying no arbitrage constraint

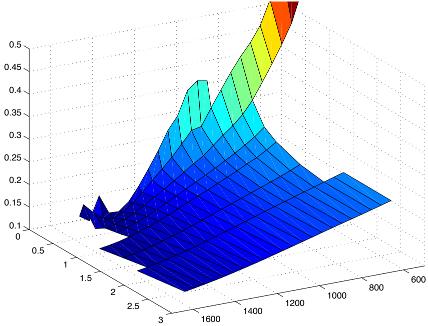
$$(S - Ke^{-r(T-t)})^+ < C < S$$

The solution $I(C)$ of equation $C_{BS}(\sigma) = C$, with C the observed price of a European call option is called *implied volatility* of this option.

Implied volatility

The implied volatility observed in the market:

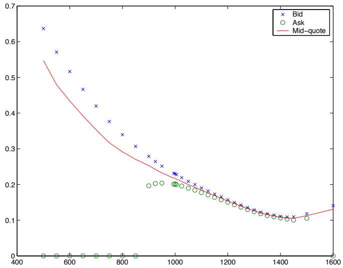
- Is often greater than the historical one;
- Depends on strike and maturity of the option
- For almost all strikes, decreases as function of strike (*implied volatility skew*).



- For very large strikes, sometimes an increase is observed (*implied volatility smile*).
- Skew and smile are more pronounced for short maturity options; the smile flattens for longer maturities.

Implied volatility

- Heding an option is more expensive in reality than in the model because of transaction costs and the need to cover sources of risk not taken into account by the model (e.g. volatility risk).
- The skew is due to the fact that the Black-Scholes model underestimates the probability of a stock market crash or a large price movement.



- The smile can be explained by liquidity premia, higher for options away from the money.

Implied volatility

Denoting $\tau = T - t$ time to maturity and $k = \log \frac{K}{S} - r\tau$ the adjusted log-strike, the implied volatility is the solution of

$$c^{BS}(k,\tau),\sigma) := N\left(\frac{-k + \frac{\sigma^2\tau}{2}}{\sigma\sqrt{\tau}}\right) - e^k N\left(\frac{-k - \frac{\sigma^2\tau}{2}}{\sigma\sqrt{\tau}}\right) = \frac{C^{obs}(\tau,k)}{S} := c^{obs}(\tau,k).$$

The ATMF implied volatility satisfies a simplified equation:

$$N\left(\frac{\sigma\sqrt{\tau}}{2}\right) - N\left(-\frac{\sigma\sqrt{\tau}}{2}\right) = c^{obs}(\tau,0).$$

When $\tau \to 0$, the ATMF implied volatility satisfies

$$I(\tau,0) \sim c^{obs}(\tau,0)\sqrt{\frac{2\pi}{\tau}}$$

A key property of implied volatility surface, is the *implied volatility skew*, defined by $\partial_k I(\tau,k)$.

Implied volatility skew

- Differentiating the Black-Scholes formula:

$$\partial_k I(\tau,k) = \frac{\partial_k c(k,\tau) - \partial_k c^{BS}(k,\tau,I(\tau,k))}{\partial_I c^{BS}(k,\tau,I(\tau,k))}$$

$$= \frac{\mathbb{P}^{BS}[X_\tau \geq k] - \mathbb{P}[X_\tau \geq k]}{e^{-k} n(d_1)\sqrt{\tau}},$$

  where under $\mathbb{P}^{BS}$, S_τ follows the BS model with volatility $I(\tau,k)$.
- Substituting $k=0$, we obtain the formula for ATMF skew:

$$\partial_k I(\tau)_{ATMF} = \frac{\mathbb{P}^{BS}[X_\tau \geq 0] - \mathbb{P}[X_\tau \geq 0]}{n\left(\frac{I(\tau)\sqrt{\tau}}{2}\right)\sqrt{\tau}}.$$

- As $\tau \to 0$, under above conditions,

$$\partial_k I(\tau)|_{ATMF} = \frac{\sqrt{2\pi}}{\sqrt{\tau}}\left\{\mathbb{P}^{BS}[X_\tau \geq 0] - \mathbb{P}[X_\tau \geq 0]\right\}\{1 + O(I^2(\tau,0)\tau)\}$$

- ATM skew: measure of *asymmetry of the stock price distribution function around the money*, compared to the Black-Scholes model.

Example: shifted log-normal dynamics

Let $r = q = 0$ and suppose that the process S follows the "shifted log-normal" model:

$$dS_t = (S_t - \beta)\sigma dW_t,$$

where $\beta < S$ and W is a standard Brownian motion under $\mathbb{Q}$:

$$S_t = \beta + (S - \beta)e^{\sigma W_\tau - \frac{\sigma^2 \tau}{2}}$$

The ATMF option price satistifes

$$c(0,\tau) = \frac{S-\beta}{S}\mathbb{E}[(e^{\sigma W_t - \frac{\sigma^2 t}{2}} - 1)^+] = \frac{S-\beta}{S}c^{BS}(0,\tau,\sigma).$$

so that, as $T \to 0$,

$$I(0,T) \sim \frac{S-\beta}{S}c^{BS}(0,T,\sigma)\sqrt{\frac{2\pi}{T}} \to \frac{S-\beta}{S}\sigma.$$

Shifted log-normal dynamics: ATM skew

We have:

$$\mathbb{P}^{BS}[X_\tau \geq 0] = \mathbb{P}[e^{\sigma^* W_\tau - \frac{(\sigma^*)^2 \tau}{2}} \geq 1] = N(-\sigma^* \sqrt{\tau}/2);$$

$$\mathbb{P}[X_\tau \geq 0] = \mathbb{P}[\beta + (S_0 - \beta)e^{\sigma W_\tau - \frac{\sigma^2 \tau}{2}} \geq S] = \mathbb{P}[e^{\sigma W_\tau - \frac{\sigma^2 \tau}{2}} \geq 1] = N(-\sigma\sqrt{\tau}/2).$$

Therefore,

$$\partial_k I(\tau) = \frac{\sqrt{2\pi}}{\sqrt{\tau}}\{N(-\sigma^*\sqrt{\tau}/2) - N(-\sigma\sqrt{\tau}/2)\} \to \frac{\sigma\beta}{2S}$$

as $\tau \to 0$.

$\Rightarrow$ the implied volatility skew has the same sign as β.